

CryptoShield

Next Frontier of Protection

Intelligent Hybrid File Encryption System with Automatic Cryptographic
Algorithm Recommendation



The Title

CryptoShield: Intelligent Hybrid File Encryption System
with Automatic Cryptographic Algorithm
Recommendation

01

Techniques

AES-256-GCM, Triple DES, RSA-2048,
SHA-512

02

Domain

Cryptography & Cybersecurity

03

Subtitle

Secure File Encryption, Integrity
Verification and Adaptive
Cryptographic Algorithm Selection

Objectives

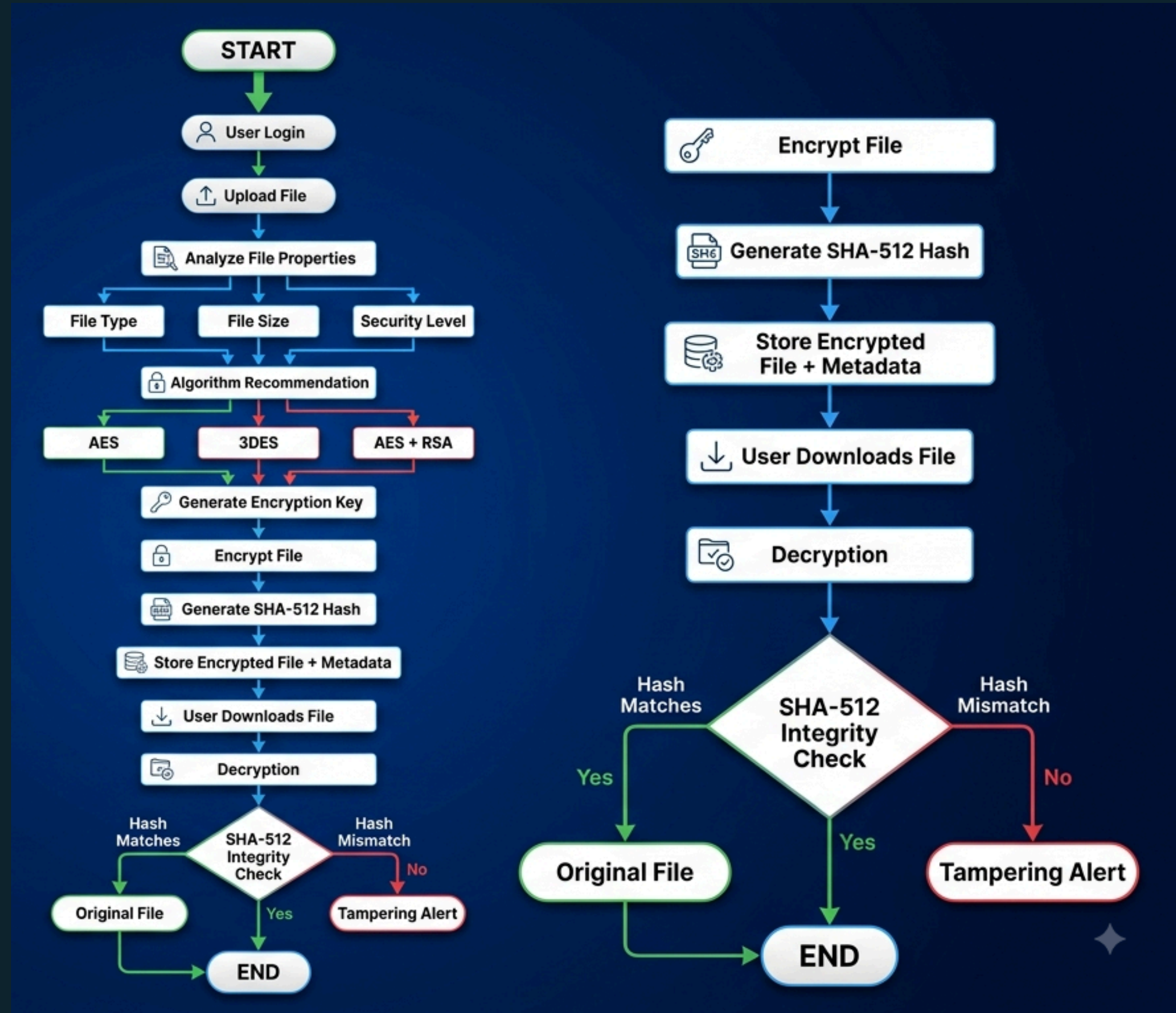
Primary Objective

- To develop a secure file protection system that automatically recommends an appropriate cryptographic algorithm based on file type, file size and required security level.

Specific Objectives

- Implement AES-256-GCM for secure file encryption and decryption.
- Use RSA-2048 for secure encryption/protection of AES session keys.
- Implement SHA-512 for file integrity verification and tamper detection.
- Include Triple DES for comparative analysis with modern encryption techniques.
- Analyze encryption/decryption time, file size and performance.
- Provide a user-friendly web interface for uploading, encrypting, decrypting and verifying files.
- Develop an adaptive recommendation mechanism for selecting a suitable encryption strategy.

Flow Chart



DataSet

Self-Generated Cryptographic Benchmark Dataset

- File Types: PDF, DOCX, TXT, JPG, PNG, ZIP, MP4
- File Sizes: Small, Medium, Large
- Security Levels: Low, Medium, High
 - Parameters Recorded: File size
 - Encryption algorithm
 - Encryption time
 - Decryption time
 - SHA-512 hashing time
 - Encrypted file size

Purpose:

To evaluate and compare the performance of AES, Triple DES, and hybrid AES-RSA encryption.

Downloadable Source

For creating the test files, we can use public datasets from:

<https://www.kaggle.com/datasets/chaitanya205/cryptographic-algorithm-classification-dataset>

Methodology Overview

- File Upload: User uploads a file and selects the required security level.
- File Analysis: The system analyzes the file type and size.
- Algorithm Recommendation: The system recommends a suitable encryption method based on the file characteristics.
- Encryption: The selected algorithm encrypts the file securely.
- Integrity Verification: SHA-512 generates a hash to detect file tampering.
- Decryption: The encrypted file is decrypted back to its original form.
- Performance Analysis: Encryption and decryption time are measured and compared.

Thank You